Aria-NBV: Master's Thesis Outlook

Supervisor agenda: scope lock, evidence bar, and geometry-first non-myopic planning

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Master's Thesis Proposal

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Meeting: Top 7 Items

Discuss In This Order

1. **Compute? LRZ / workstation** first; if local, keep the stack in **WSL**.
2. **Access?** Apply now for **Aria Gen2** + **ASE simulator**.
3. **Core?** Stay on the **ASE / mesh-backed** stack and prove **geometry-first non-myopic planning**.
4. **Ablation budget?** Do the **minimum** single-step **VIN** work needed, then move to multi-step.
5. **State contract? Geometry-only counterfactuals** by default; **RGB / 3DGS / simulator** later.
6. **Planning target?** Pick the **return proxy**, method family, and **discrete vs continuous** phase-1 action space.
7. **Hierarchy scope?** Keep **VLA / semantic-global planning** as phase 2, not thesis core.

Decision 1: VIN / Ablation Budget

Topic

1. single-step **VIN** only matters if it can become a trusted **reward / critic** beyond the mesh subset
2. this decision fixes how much **EVL / EFM backbone** and **CORAL** work is actually worth doing now

Options

1. **A. Minimum:** fix calibration, ablate **CORAL** / aux loss / key inputs, then move to multi-step.
2. **B. Architecture-first:** do heavier **VIN-v4** / backbone search before multi-step planning.

Questions:

1. How much **single-step VIN** work is worth doing before multi-step?
2. Do we only need **trust-building** ablations, or a stronger model that can later provide **reward / critic** estimates beyond the mesh subset?
3. Is the current **EVL / EFM** backbone already sufficient, including a **CORAL / binning** sanity ablation?

Recommendation: choose **A**: do the **minimum** single-step work needed to trust the reward model, then move to multi-step

Decision 2: Counterfactual State Contract

Topic

1. What modalities should **history** and **counterfactual** state actually use?
2. This decides whether multi-step stays geometry-first or depends on expensive RGB synthesis.

Options

1. **Geometry-only everywhere**
2. **Full history modalities + geometry-only counterfactuals**
3. **Counterfactual RGB / semantics** via **3DGS** or simulator

Questions:

1. Do we need counterfactual **SLAM PC** emulation, or is $\mathcal{P}_t$ semi-dense plus $\mathcal{P}_q$ dense already sufficient?
2. Should history use **all logged modalities**, while counterfactual state stays **geometry-only** by default?

Recommendation: choose **B** now; only escalate to **C** if geometry-only counterfactuals clearly fail

Decision 3: Planning Formulation

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Topic

1. phase 1 should lock the **multi-step target**, **method family**, and **action space** before worrying about online refinement

Options

1. **A. Search / discrete-shell first:** beam, MPC, close-greedy, or offline RL over the shell
2. **B. Actor-critic / continuous early:** learn continuous actions and feasibility together from the start

Questions:

1. What is the right **multi-step return proxy**: direct oracle RRI sum, discounted **VIN**-predicted reward, or a mixed oracle / surrogate target?
2. Which methods fit our setting best: **beam / MPC / close-greedy**, **IQL / CQL**, or actor-critic RL?
3. Should phase 1 stay **discrete-shell**, or already include **continuous** actions?
4. Should candidate rules be handled only by masks / reward terms, or should **VIN** also predict feasibility / rule violations?

Recommendation: choose **A** now; lock the return proxy and method family on the discrete shell before moving toward continuous control

Decision 4: RL Regime + Data Collection

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Topic

1. online RL becomes realistic only once reward evaluation and counterfactual rendering are cheap enough

Options

1. **A. Offline first:** search, close-greedy, discrete-shell RL, and modality-contract work
2. **B. Offline + online:** add simulator-based data collection and policy refinement early

Questions:

1. Should we prioritize **offline only** first, or **offline + online** RL together?
2. Should early RL effort focus on **modality contract + return proxy** before simulator-based data collection?

Recommendation: choose **A** now; use oracle / VIN rewards offline first, and keep online RL conditional on simulator access plus a fast surrogate reward

Decision 5: Hierarchical Scope

Topic

1. hierarchy can mean either **local geometric target-selection** or a larger **semantic-global planner**

Options

1. **A. Local hierarchy only:** target / look-at prediction plus local viewpoint realization
2. **B. Semantic-global hierarchy: VLA / VLM** chooses portals, entities, or frontiers for the local controller

Questions:

1. Should **semantic-global planning** via **VLA / VLM** be part of thesis core, or stay a phase-2 extension?
2. If hierarchy is pursued now, should it stay **local geometric** rather than full semantic-global planning?

Recommendation: choose **A** for thesis core; keep **VLA / semantic-global planning** explicitly out of phase 1

Theory: Geometry-First MDP Contract

Historical vs Counterfactual State

- use the **logged ego history** we already have, but keep the **implemented counterfactual view** geometry-safe
- phase-1 state can stay **current pose + visited poses + shell candidates + valid mask**; no real RGB, SLAM, or semantics at unvisited poses unless we synthesize them
- rollout trees and RL episodes already instantiate this contract, but the canonical offline dataset is still **one-step**

$$s_t = \{\mathcal{O}^{\text{ego}}_{1:t}, \mathcal{P}_t, \mathcal{Q}\}$$

$$s^{\text{cf}}_t = \left\{\left(\mathcal{O}^{\text{ego}}, \mathcal{O}^{\text{cf}}\right)_{1:t}, \mathcal{P}_t, \mathcal{Q}\right\}$$

Action, Reward, Return Proxy

- today the **implemented** action is a **discrete shell slot**; phase 2 can factor it into target z_t plus pose $\boldsymbol{x}_t$, reusing the same env contract (`counterfactual_env.py`)
- default reward is **oracle** RRI, **not** VIN; VIN remains the future surrogate / critic hook
- the current regime is **short-horizon / close-greedy**; the clean next step is to persist `top-k` chains plus discounted return into `vin_offline.counterfactuals`

$$a_t \in \{0, \ldots, N_{\text{shell}} - 1\}, \quad q_t = \text{shell}[a_t]$$

$$a^{\text{phase2}}_t = \{z_t, \boldsymbol{x}_t\}, \quad r_t = \text{RRI}(q_t)$$

$$G^H_t = \sum_{k=0}^{H-1} \gamma^k r_{t+k}, \quad \gamma = 0.1$$

Theory Outlook: Reward / Q / Critic Boundary

Keywords + Equations

- **oracle RRI reward** now; **VIN surrogate / critic** later
- **hard masks** already handle collision, clearance, and bounds
- **low-discount return** keeps phase 1 close-greedy
- **Q / critic** should estimate cumulative quality, not coverage

$$r_t^{\text{geom}} = \log(\text{CD}(\mathcal{P}_t, \mathcal{M}_{\text{GT}}) + \varepsilon) - \log(\text{CD}(\mathcal{P}_{t+1}, \mathcal{M}_{\text{GT}}) + \varepsilon) - \alpha \mathbb{1}[\text{collision}(a_t)] - \beta c(a_t)$$

$$y_t^Q = r_t + \gamma V(s_{t+1})$$

Implemented Now

- candidate rules already reject **too-close**, **collision**, and **out-of-bounds** actions (`candidate_generation_rules.py`)
- the RL env already uses **oracle-RRI immediate reward** plus a flat invalid-action penalty (`counterfactual_env.py`)
- cumulative rollout RRI and low-γ PPO diagnostics already exist (`counterfactuals.py`)

Needs Change

- make rule penalties **explicit in reward**
- add a **VIN-backed counterfactual evaluator**
- decide whether a **privileged critic** may use GT mesh / OBB cues

Hestia: What Actually Transfers

Keep These Ideas

- **directional observability:** encode viewing direction, not only seen / unseen
- **hierarchical factorization:** split `_where to attend_` from `_where to move_`
- **supervised target head:** train target / attention separately
- **close-greedy control:** start with small gamma / short horizons

What We Should Not Borrow

- do not replace RRI with coverage reward
- move toward **target-conditioned local reads**, not fixed shell scoring forever
- do not jump straight to monolithic continuous 5DoF PPO
- keep hierarchy as a **phase-2 extension**; use Hestia as **planning decomposition**

VINv4 Bridge: From Shell Scoring To Target Control

What Changes Beyond Fixed Candidate Scoring

- predict an intermediate target latent z_t before the pose
- supervise z_t from expected RRI gain, uncertainty, or entity deficit
- read local EVL / geometry **at the target**, not only at fixed shell poses
- bridge from discrete shell scoring toward later continuous control

$$a_t^{\text{phase2}} = \{z_t, \boldsymbol{x}_t\}$$

What Stays The Same

- keep the same quality objective: RRI now, $\hat{r}_t$ later
- keep EFM3D / EVL geometry priors as the backbone
- treat the target head as a **bridge**, not a replacement for the shell baseline
- route targets through OBBs / SceneScript entities for **entity-aware** RRI

Implemented: Rollout Scaffold

What's done?

- rollouts already reuse the **same candidate-shell contract** as one-step NBV (`counterfactuals.py`)
- the oracle scorer already computes **incremental** and **cumulative** RRI
- horizon, branching, pruning, distance guards, and plotting already exist (`plotting.py`)
- this is a real scaffold, not a blank start

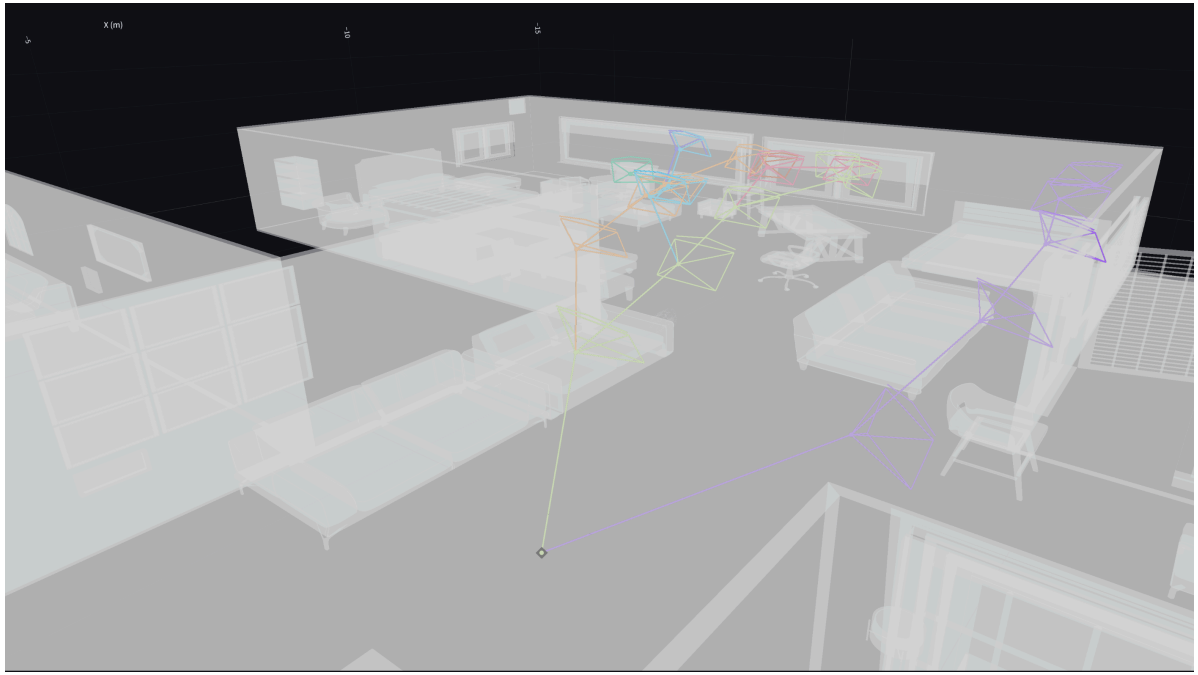

Figure 1: Synthetic diagnostic only: multi-step counterfactual rollout tree from the implemented `app/multi-step` plotting surface.

Implemented: RL Inspector + Step Diagnostics

What's done?

- the Gymnasium RL env already exists on the discrete shell, with greedy / random baselines and SB3 PPO smoke coverage (`counterfactual_env.py`)
- plotting already supports shell inspection, selected frusta, and trajectory replay (`plotting.py`)
- the RL page is evaluation-first: shell preview, episode replay, and policy comparison (`rl.py`)
- greedy immediate reward already beats random; next step = evidence

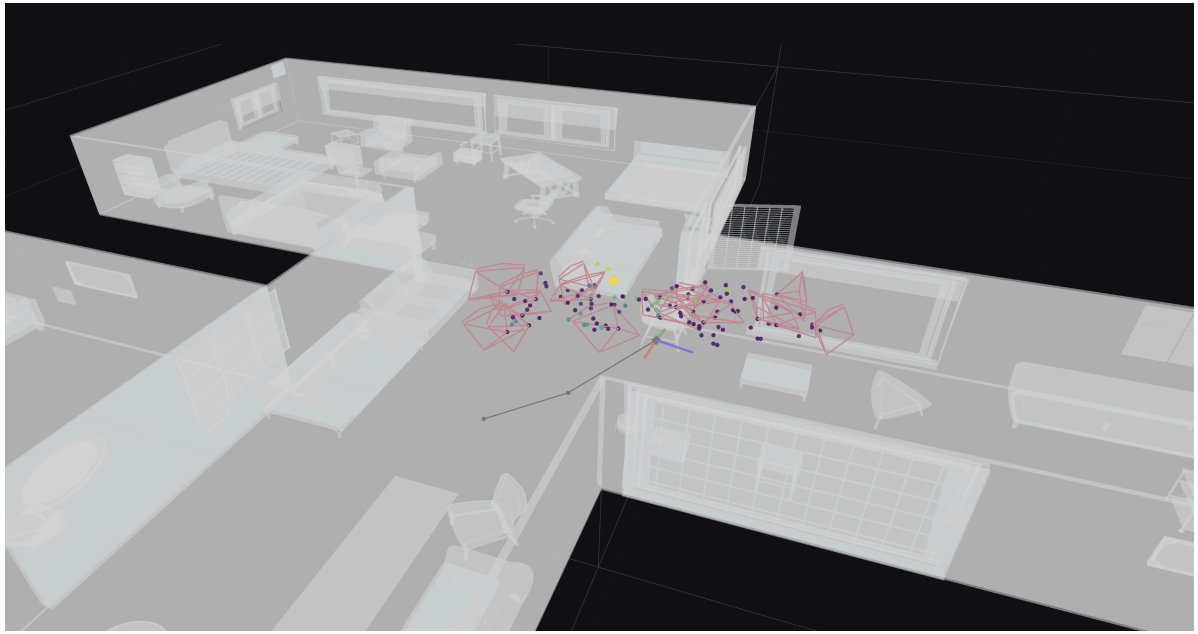

Figure 2: Synthetic diagnostic only: step-level shell / frusta view from the same `app/multi-step` figure family.

Roadmap: Thesis Core vs Phase-2 Extensions

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VINv4 / Hierarchy Execution Path

1. directional bookkeeping on EVL / semidense state
2. supervised target head from expected RRI gain / uncertainty / entity deficit
3. target-aware local interpolation / feature reads
4. target-conditioned controller plus feasibility projection
5. close-greedy training before long-horizon RL

Scaling + Entity-Aware Extensions

- scale the mesh-backed subset, anchor poses, and candidate bounds
- compare **beam / MPC / close-greedy** against one-step greedy
- route target selection through OBBs / SceneScript entities for **entity-aware** RRI
- only after that: stronger critics, counterfactual RGB / 3DGS, and semantic-global planning

Desired Thesis Deliverable

- a clear **geometry-first MDP**, a defensible **non-myopic baseline**, and an honest scope statement

Open Questions After Scope Lock

Still Open Scientifically

- how much counterfactual history should the actor observe vs critic only?
- can a privileged critic use GT mesh / OBB / segmentation cues?
- how do we scale beyond mesh-backed scenes?
- which anchor poses and candidate bounds define the real training distribution?

Still Open Operationally

- when do we broaden beyond 60 candidates and many more anchor poses?
- how much effort should go into VIN v4 once the baseline exists?
- when, if ever, do RGB / 3DGS move into thesis core?
- what is the minimum convincing experiment set?

Backup: Simulator Options If Needed

Most Relevant Candidates

1. **ASE simulator / generation stack:** best fit if accessible
2. **Habitat-Sim:** best fast geometry-first sidecar
3. **AI2-THOR / ProcTHOR:** useful for semantic-global planning, not local NBV sensing
4. **Isaac Sim:** best public one-box multimodal candidate

Recommendation

- this is mainly a **data-contract** question, not just a renderer choice
- near-term, do **not** switch ecosystems before the geometry-first baseline is proven
- if public tooling is needed: **Isaac Sim** main, **Habitat-Sim** sidecar
- any choice still must satisfy Aria optics, RGB / SLAM streams, GT geometry, OBB / semantics, and a plausible semi-dense / EVL replacement